
Software Requirements Specification

Project: NeuroForge Enterprise SDLC Platform

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the requirements of the NeuroForge Enterprise SDLC Platform.

The platform is designed to help software development teams manage software projects and different activities of the Software Development Life Cycle (SDLC) in an organized manner. This document describes the main features, users, interfaces, and requirements of the system.

1.2 Intended Audience

This document is intended for:

- Project team members
- Project mentors and reviewers
- Software developers
- Testers
- Project managers
- Other stakeholders involved in the project

1.3 Scope

The NeuroForge Enterprise SDLC Platform is a web-based platform for managing software development activities.

The system aims to provide a common platform for managing:

- Software projects
- Project requirements
- Tasks
- Team members
- Project progress
- Testing activities

The platform helps development teams organize their work and manage project activities more efficiently.

1.4 Definitions

Term	Definition
SDLC	Software Development Life Cycle
SRS	Software Requirements Specification
Admin	A user who manages the system and users
Project Manager	A person who manages project activities
Team Member	A person who works on assigned tasks
Tester	A person who performs testing activities
Requirement	A feature or function expected from the system

1.5 References

The following references may be used for preparing the requirements:

- Project discussions and team requirements
 - Infosys Springboard project guidelines
 - Software Requirements Specification guidelines
 - Software Development Life Cycle concepts
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2. Overall Description

2.1 User Interface

The NeuroForge platform will provide a simple and user-friendly interface.

The user interface may include:

- Login page
- Dashboard
- Project management page
- Requirements page
- Task management page
- Testing page
- Reports page

The interface should allow users to navigate easily and access the features according to their roles.

2.2 System Interface

The system will interact with:

- Users through a web browser
- A database for storing system information
- The backend system for processing user requests
- Other required services or systems used by the platform

The system will receive user input, process the information, and display the required results to the user.

2.3 Software and Hardware Requirements

Software Requirements

The system may require:

- Operating System: Windows, Linux, or macOS
- Web Browser: Google Chrome, Microsoft Edge, or Mozilla Firefox
- Database Management System
- Required programming language and development tools
- Internet connection

Hardware Requirements

The system can be accessed using:

- Computer or laptop
- Minimum 4 GB RAM
- Sufficient storage space
- Keyboard and mouse
- Internet connection

The exact software and hardware requirements may depend on the technologies selected for the project.

2.4 Constraints

The system may have the following constraints:

- Users must have valid login credentials to access protected features.
- Users can access features based on their assigned roles.
- The system requires internet connectivity when hosted online.
- The system must follow the technologies and design decisions selected by the project team.
- The system should be developed within the available time and resources.

2.5 User Characteristics

The system may be used by different types of users.

Administrator

The Administrator manages users and controls system access.

Project Manager

The Project Manager manages projects, requirements, tasks, and project progress.

Team Member

The Team Member works on assigned tasks and updates task status.

Tester

The Tester performs testing activities and records testing information.

Users are expected to have basic knowledge of using web applications and their assigned project responsibilities.

3. System Features and Requirements

3.1 Functional Requirements

FR-01: User Login

The system shall allow registered users to log in using valid login credentials.

FR-02: User Management

The system shall allow the Administrator to manage user accounts.

FR-03: Role-Based Access

The system shall provide access to system features based on the user's role.

FR-04: Project Management

The system shall allow authorized users to create, view, and update project information.

FR-05: Requirement Management

The system shall allow users to add, view, update, and manage project requirements.

FR-06: Task Management

The system shall allow authorized users to create and manage project tasks.

FR-07: Task Assignment

The system shall allow Project Managers to assign tasks to Team Members.

FR-08: Task Status Update

The system shall allow Team Members to update the status of their assigned tasks.

FR-09: Project Progress Tracking

The system shall allow users to view the progress of projects and tasks.

FR-10: Testing Management

The system shall allow Testers to record and track testing activities.

FR-11: Report Generation

The system shall allow authorized users to view project-related reports.

FR-12: User Logout

The system shall allow users to log out of the system securely.

3.3 External Interface Requirements

User Interface

The system should provide a simple interface through which users can:

- Log in
- View their dashboard
- Manage projects
- Manage requirements
- View and update tasks
- Track progress
- Access reports

Hardware Interface

The system should be accessible through computers or other suitable devices connected to the internet.

Software Interface

The system may interact with:

- Web browsers
- Database systems
- Backend services
- Other software components required by the project

Communication Interface

The system requires network communication for communication between the user interface, backend system, and database.

3.4 Database Requirements

The system should store and manage the following information:

User Information

- User ID
- Name
- Email
- Role
- Login information

Project Information

- Project ID
- Project name
- Project description
- Project status

Requirement Information

- Requirement ID
- Requirement description
- Requirement status
- Related project

Task Information

- Task ID
- Task description
- Assigned user
- Task status
- Related project

Testing Information

- Test details
- Test status
- Testing results
- Related project or requirement

The database should store information in an organized manner and allow authorized users to access the required data.

3.5 Non-Functional Requirements

Usability

The system should be simple and easy to use. Users should be able to navigate through the platform without difficulty.

Performance

The system should respond to user requests within a reasonable time.

Security

Only authorized users should be allowed to access protected information and system features.

Reliability

The system should perform its functions correctly and consistently.

Maintainability

The system should be easy to update and maintain when changes are required.

Availability

The system should be available to users when the required system and network services are available.

Scalability

The system should be capable of supporting additional users, projects, and data as the system grows.